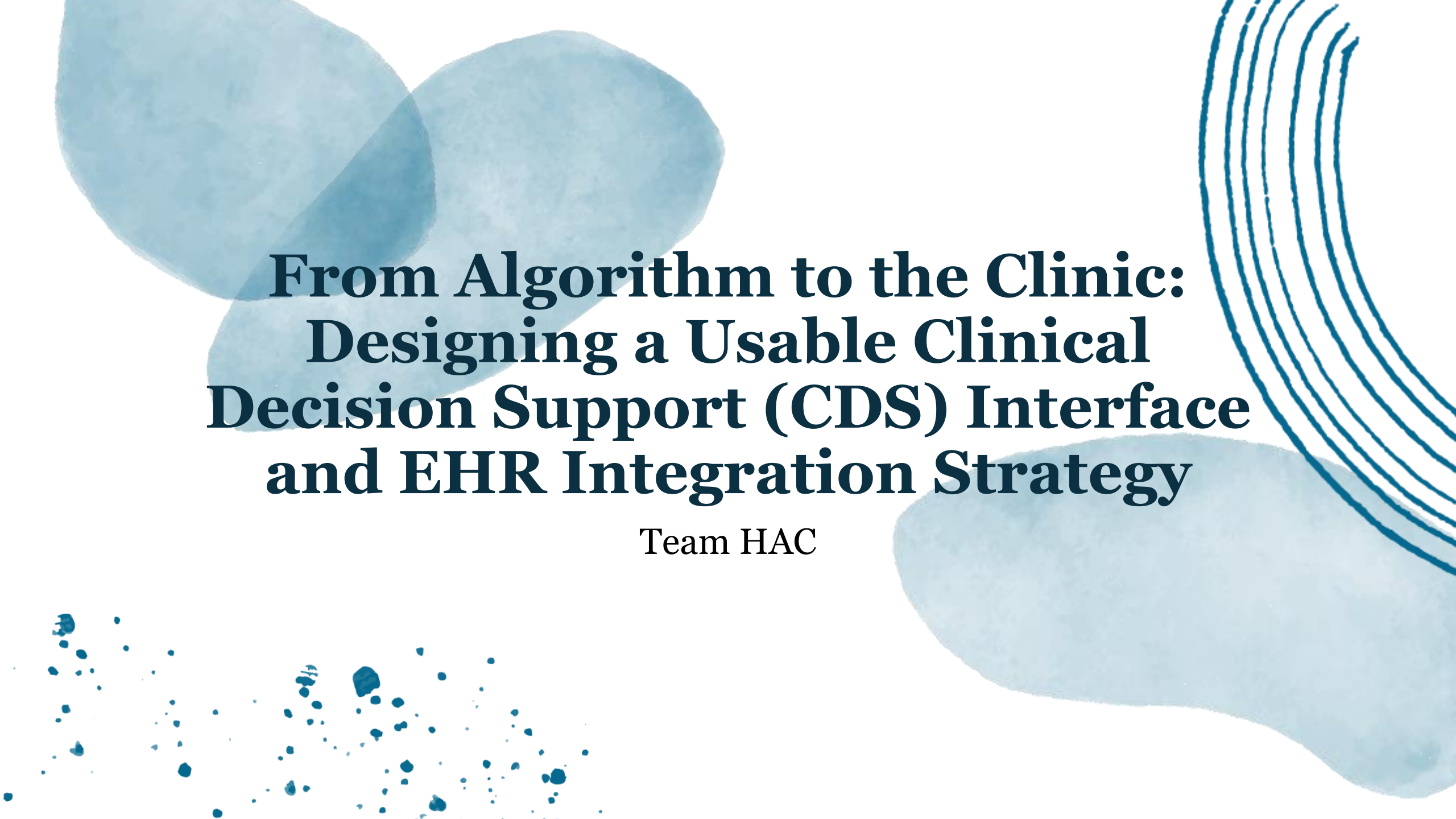
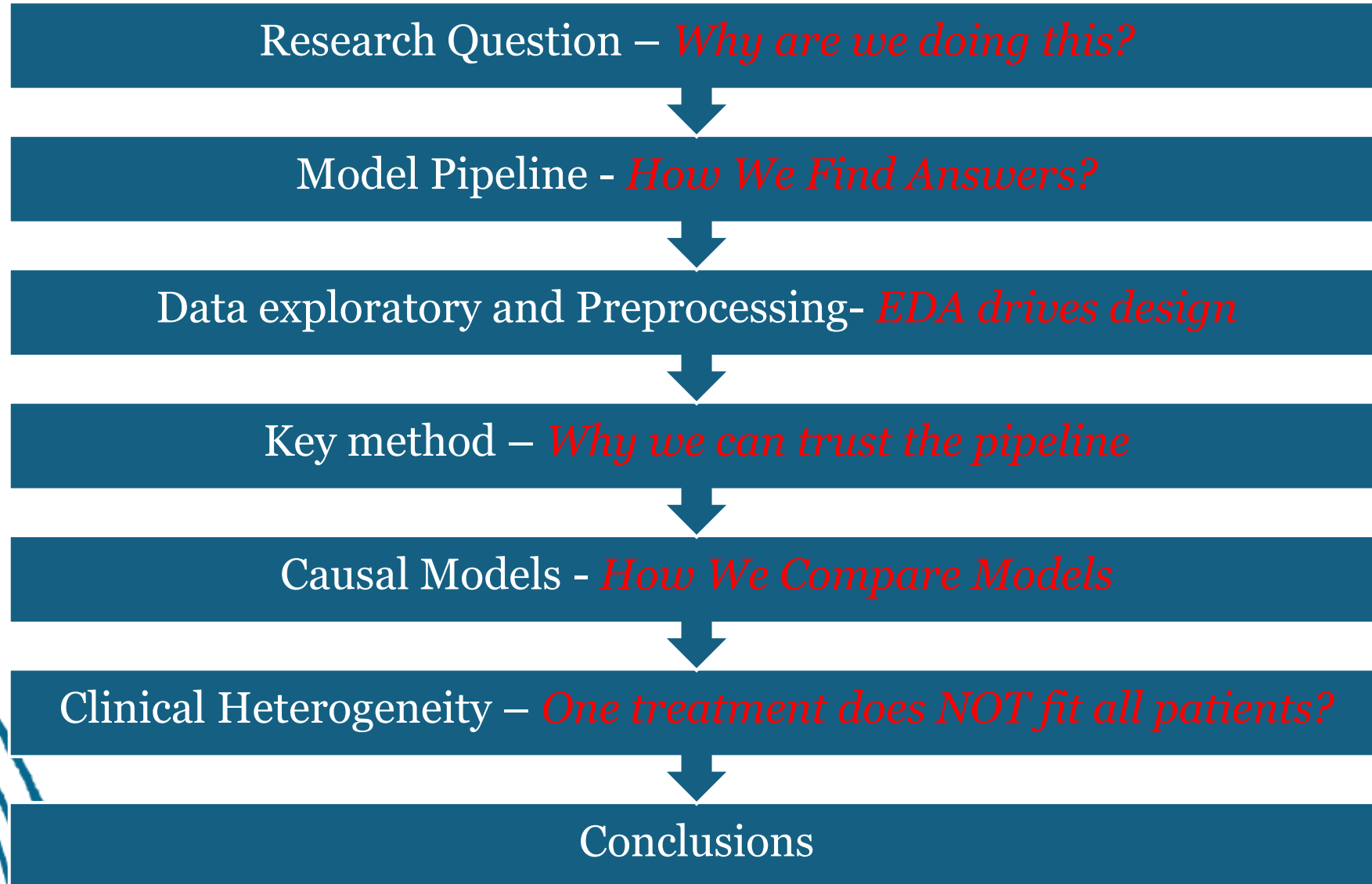




From Algorithm to the Clinic: Designing a Usable Clinical Decision Support (CDS) Interface and EHR Integration Strategy

Team HAC

Roadmap (1/1)



We would like to thank the authors of the papers referenced in this study.

- **Asgharnezhad, A., et al.** (2025). *Enhancing Monte Carlo dropout performance for uncertainty quantification* (arXiv preprint arXiv:2505.15671). <https://arxiv.org/pdf/2505.15671>
- **Austin, P. C.** (2021). *Missing data in clinical research: A tutorial on multiple imputation*. U.S. National Library of Medicine. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8499698/>
- **Djupskås, A., Stasik, A. J., & Riemer-Sørensen, S.** (2025). *Unreliable uncertainty estimates with Monte Carlo dropout* (arXiv preprint arXiv:2512.14851). <https://arxiv.org/abs/2512.14851>
- **Lee, K. M., & Emsley, R.** (2024). The impact of heterogeneity on the analysis of platform trials with normally distributed outcomes. *BMC Medical Research Methodology*, 24, Article 163. <https://doi.org/10.1186/s12874-024-02293-4>
- **Tamo, J. B.** (2026). Causal machine learning for surgical interventions. *Big Data Mining and Analytics*. <https://www.scipopen.com/article/10.26599/BDMA.2025.9020093>
- **Tian, H., Tom, B. D. M., & Burgess, S.** (2024). A data-adaptive method for investigating effect heterogeneity with high-dimensional covariates in Mendelian randomization. *BMC Medical Research Methodology*, 24, Article 34. <https://doi.org/10.1186/s12874-024-02153-1>
- **Yang, T., & Noor-E-Alam, M.** (2025). *Optimizing feature selection in causal inference: A three-stage computational framework for unbiased estimation*. arXiv. <https://arxiv.org/abs/2502.00501>
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- **Yang, J., Wan, Y., Ni, Q., et al.** (2022). Quantifying causal effects from observed data using quasi-intervention. *BMC Medical Informatics and Decision Making*, 22, 337. <https://doi.org/10.1186/s12911-022-02086-z>
- **Austin, P. C., White, I. R., Lee, D. S., & van Buuren, S.** (2021). Missing data in clinical research: A tutorial on multiple imputation. *The Canadian Journal of Cardiology*, 37(9), 1322–1331. <https://doi.org/10.1016/j.cjca.2020.11.010>
- **Zhao, Z., Zhang, Y., Harinen, T., & Yung, M.** (2022). Feature selection methods for uplift modeling and heterogeneous treatment effect. In *Proceedings of the 18th IFIP International Conference on Artificial Intelligence Applications and Innovations (AIAI)* (pp. 217–230). Springer. https://doi.org/10.1007/978-3-031-08337-2_19

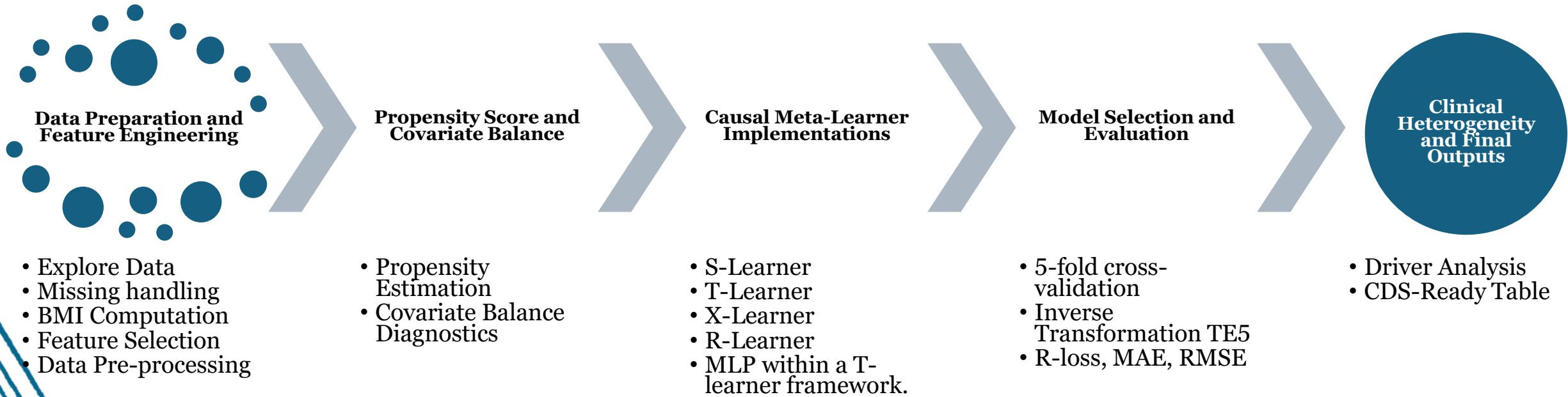


How can patient-specific treatment effects of surgery versus conservative care be estimated to support shared decision making?

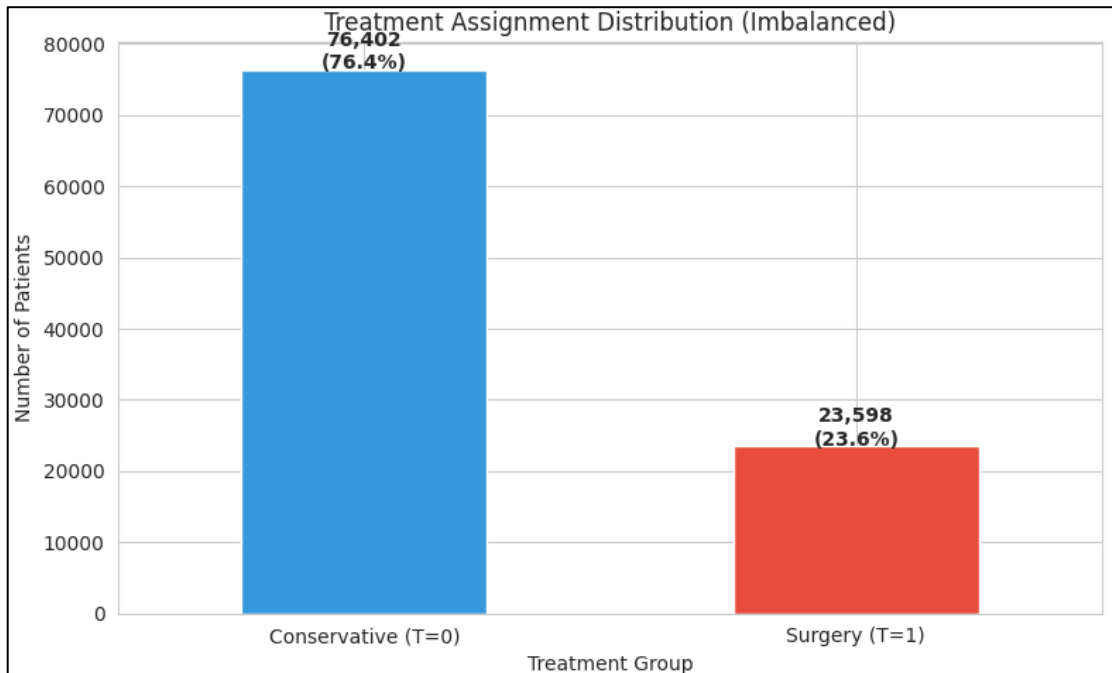
For outcome $k \in \{1, \dots, 5\}$:

$$TE_k(x) = \mathbb{E}[Y_k(1) - Y_k(0) \mid X = x]$$


Model pipeline (1/1)

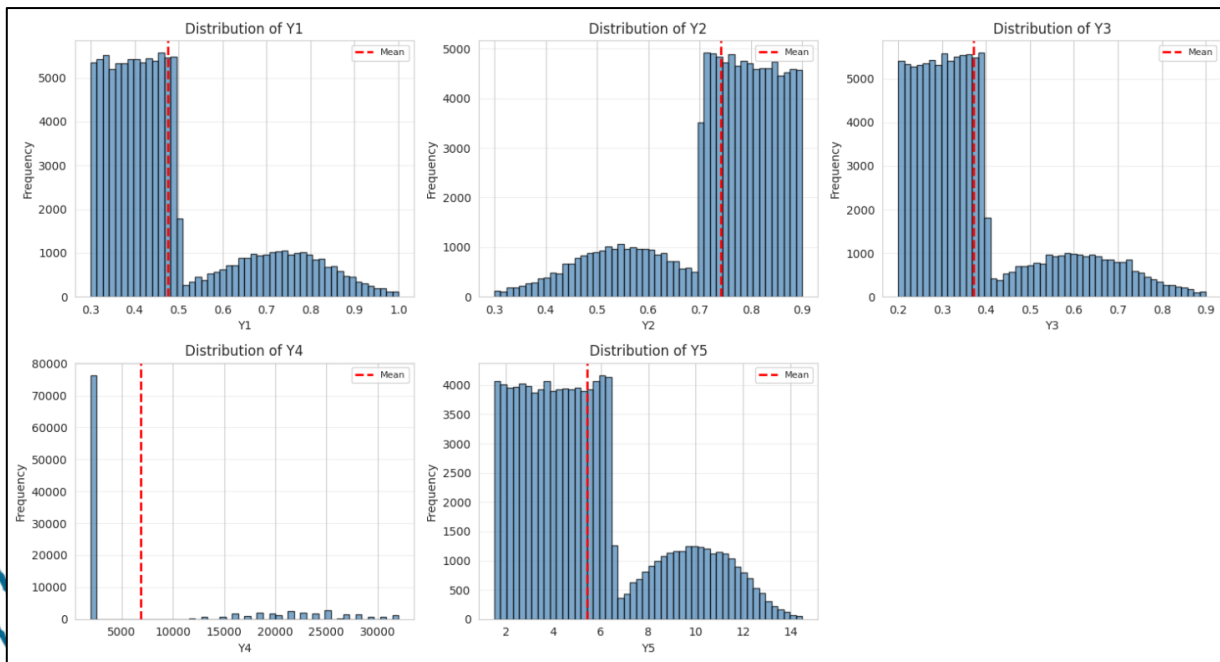


Data exploratory and Preprocessing (1/4)



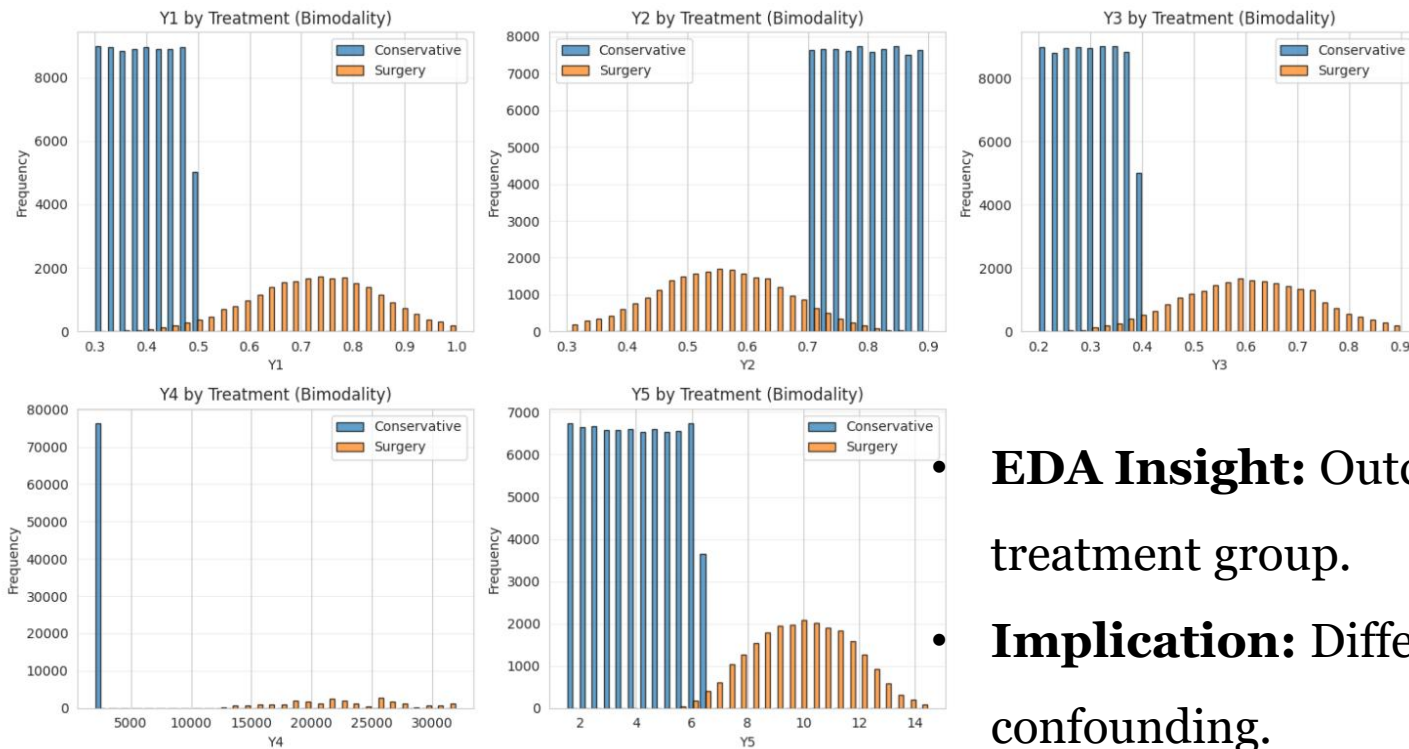
- **EDA Insight:** Treatment assignment is highly imbalanced (24% surgery).
- **Implication:** Strong selection bias; naive outcome comparisons are invalid.
- **Solution:** Propensity score modeling with IPW and balance diagnostics.

Data exploratory and Preprocessing (2/4)



- **EDA Insight:** Outcomes exhibit strong skewness and bimodality.
- **Implication:** Treatment effects are heterogeneous and non-linear.
- **Solution:**
 - Log1p transformation for cost and time outcomes.
 - Causal meta-learners with outcome-specific modeling.
 - R-loss.

Data exploratory and Preprocessing (3/4)



- **EDA Insight:** Outcome distributions differ substantially by treatment group.
- **Implication:** Differences may reflect causal effects or confounding.
- **Solution:** Doubly robust causal learning with balance diagnostics.

Data exploratory and Preprocessing (4/4)

50 columns with missing values:

Column	Missing_Count	Missing_Pct	Dtype
X51	72717	72.72	float64
X54	3081	3.08	object
X60	3081	3.08	object
X21	3069	3.07	object
X20	3055	3.06	float64
X15	3055	3.06	float64
X16	3056	3.06	object
X41	3053	3.05	object
X14	3050	3.05	datetime64[ns]
X40	3047	3.05	float64
X26	3046	3.05	float64
X36	3050	3.05	float64
X38	3043	3.04	object
X13	3042	3.04	float64
X59	3038	3.04	object
X18	3041	3.04	datetime64[ns]
X57	3043	3.04	float64
X45	3028	3.03	object
X24	3031	3.03	float64
X30	3031	3.03	float64
X35	3029	3.03	float64
X48	3019	3.02	object
X62	3022	3.02	float64
X37	3015	3.02	float64
X44	3015	3.02	object
X50	3014	3.01	object
X58	3004	3.00	float64

EDA Insight: No Duplicate, Substantial and variable missingness across baseline features.

• **Implication:** Dropping data would introduce bias and reduce power.

• **Solution:** Structured imputation with explicit missingness indicators.

Key Methodology (1/5)

Uplift-aligned F-filter

Why:

Help model focusing on who benefits more rather than just predicting outcomes.

Definition

Uplift-aligned F-filter is a feature-screening method that keeps only variables that change the treatment–control difference (individualized treatment effect).

Core formulation

For each feature X_j , compute its interaction strength with treatment:

$$F_j = \frac{\text{Var}(\mathbb{E}[Y' | X_j T = 1] - \mathbb{E}[Y' | X_j T = 0])}{\text{Residual variance}}$$

Interpretation:

- High F_j : feature changes the gap between treatment and control → keep it
- Low F_j : feature predicts outcome but not treatment effect → drop it

Key Methodology (2/5)

Propensity Score Weighting by Logistic Regression Model + Balance Diagnostics

Why:

Balance diagnostics justify causal estimation by confirming that treatment groups are comparable.

Definition

Propensity score weighting (IPW) reweights observational data so that treated, and control groups become comparable with respect to baseline clinical covariates, approximating a randomized experiment.

Core formulation

Propensity score:

$$e(X) = P(T = 1 | X)$$

Stabilized inverse probability weights:

$$w_i = \begin{cases} \frac{P(T = 1)}{e(X_i)} & \text{if } T_i = 1 \\ \frac{P(T = 0)}{1 - e(X_i)} & \text{if } T_i = 0 \end{cases}$$

Balance diagnostic (Standardized Mean Difference):

$$\text{SMD} = \frac{\bar{X}_{T=1} - \bar{X}_{T=0}}{s_{\text{pooled}}}$$

Key Methodology (3/5)

Causal Meta-Learners (S, T, X) with Tree-based models

Why:

Meta-learners translate heterogeneous clinical responses into patient-specific causal effects.

Definition

Meta-learners are flexible strategies that estimate individualized treatment effects by modeling potential outcomes under each treatment condition.

Core formulation

Potential outcome models:

$$\mu_1(x) = \mathbb{E}[Y' | T = 1, X = x], \mu_0(x) = \mathbb{E}[Y' | T = 0, X = x]$$

Individual treatment effect:

$$\tau(x) = \mu_1(x) - \mu_0(x)$$

Key Methodology (4/5)

Doubly Robust / R-Learner

Why:

Doubly robust learning protects causal estimates against residual confounding and model error.

Definition

The **R-learner** isolates treatment effect heterogeneity by residualizing both the outcome and the treatment assignment, making it robust to model misspecification.

Core objective

$$\min_{\tau(\cdot)} \mathbb{E}[(Y - \hat{m}(X) - (T - \hat{e}(X))\tau(X))^2]$$

Where:

$\hat{m}(X)$ is the outcome regression

$\hat{e}(X)$ is the propensity score

Key Methodology (5/5)

Cross-Fitting + R-Loss Model Selection

Why:

Cross-fitted R-loss ensures that the selected model best estimates causal effects, not just predictions.

Definition

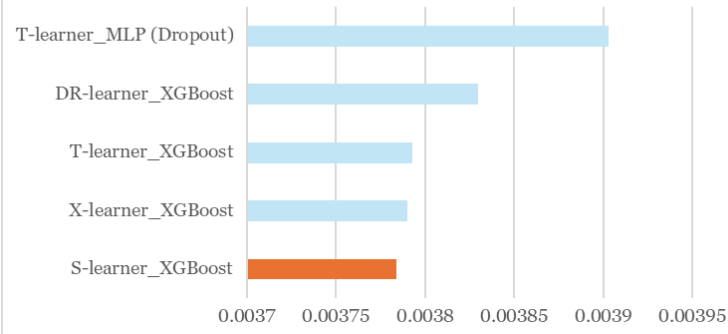
Cross-fitting separates nuisance estimation from treatment effect estimation, while **R-loss** evaluates treatment effect quality without observing counterfactuals.

Core formulation (R-loss)

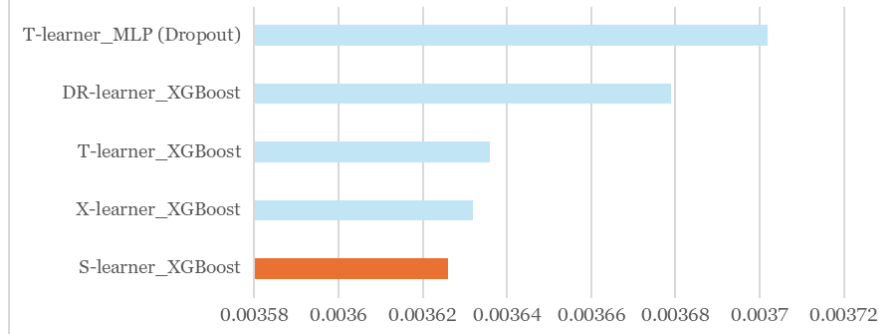
$$\mathcal{L}_R = \mathbb{E}[(Y - \hat{m}(X) - (T - \hat{e}(X))\hat{\tau}(X))^2]$$

Causal Models (1/2)

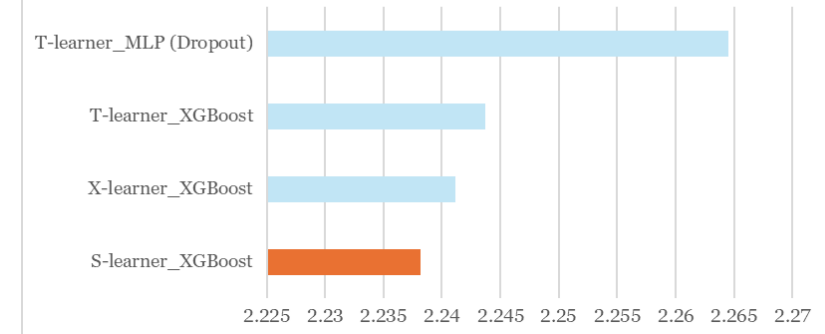
Treatment Effect Models_R-loss
Long-term pain reduction
(MLP = Multi-layer perception)



Treatment Effect Models_R-loss
Short-term treatment-related pain
(MLP = Multi-layer perception)

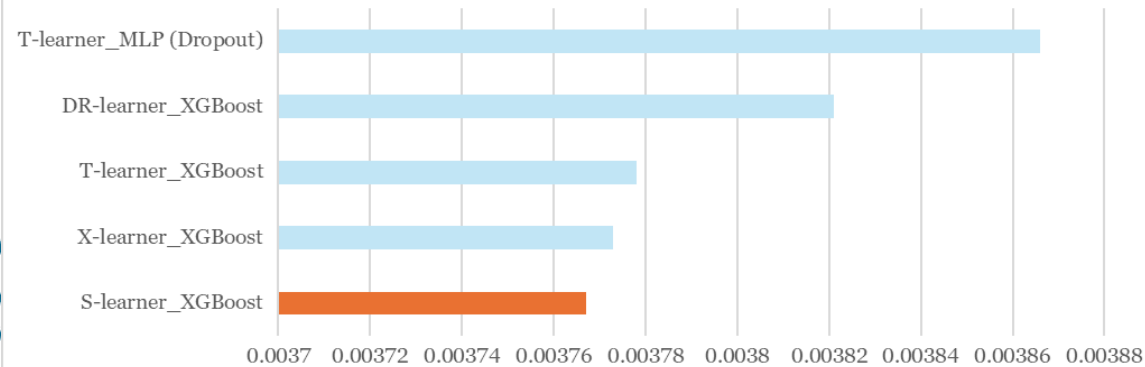


Treatment Effect Models_R-loss
Rehabilitation time (weeks)
(MLP = Multi-layer perception)

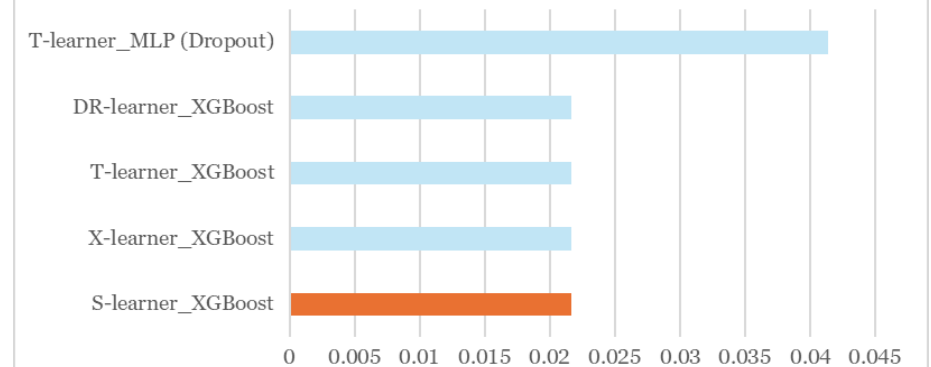


S-learner gives the most accurate individualized treatment effects

Treatment Effect Models_R-loss
Long-term functional improvement
(MLP = Multi-layer perception)



Treatment Effect Models_R-loss
Out-of-pocket costs (Dollars)
(MLP = Multi-layer perception)



Causal Models (2/2)

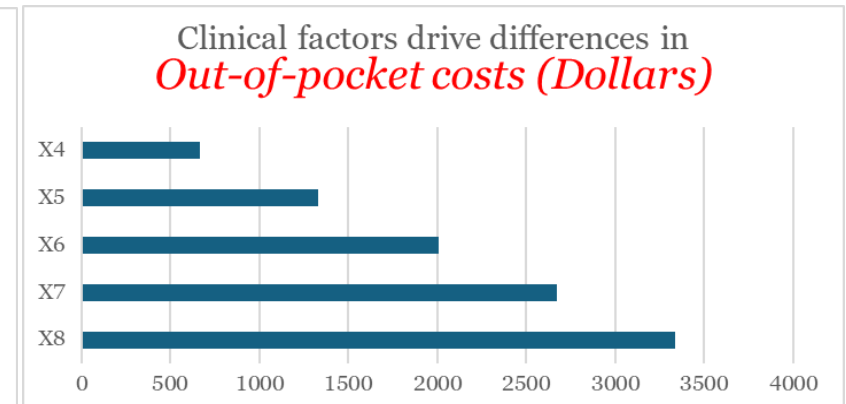
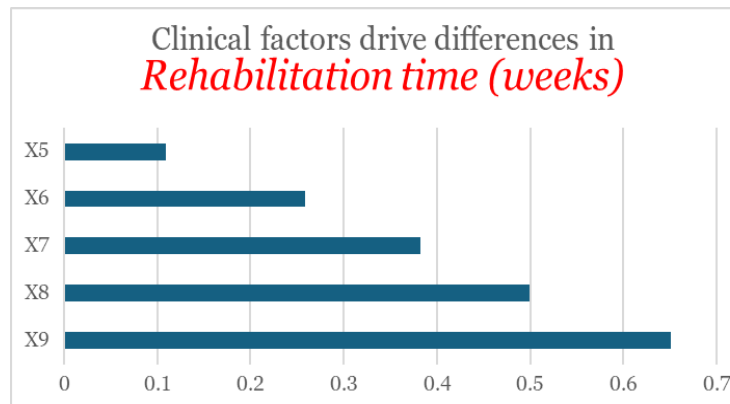
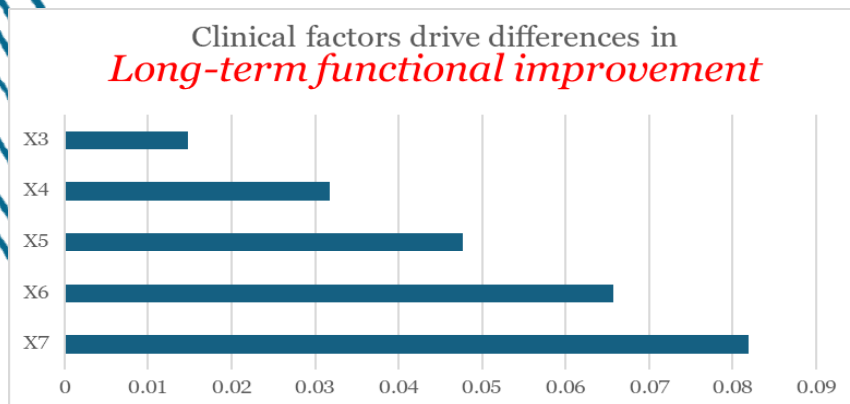
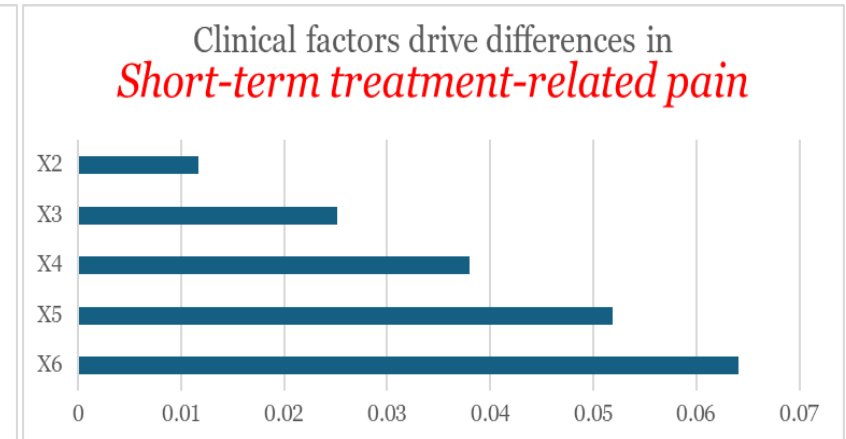
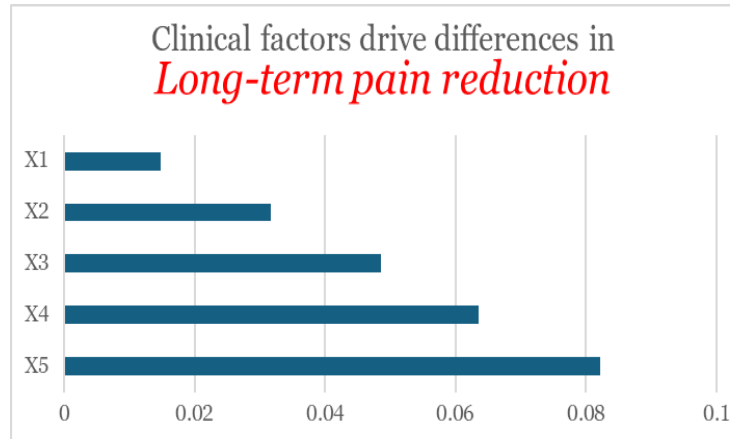
Outcome	Arm	MAE	RMSE
Y1	T=1	0.050	0.058
Y1	T=0	0.050	0.058
Y2	T=1	0.050	0.058
Y2	T=0	0.050	0.058
Y3	T=1	0.050	0.058
Y3	T=0	0.050	0.058
Y4	T=1	25.325	29.446
Y4	T=0	25.031	28.916
Y5	T=1	1.254	1.449
Y5	T=0	1.257	1.451

- **Model accuracy is balanced across treatment arms**
- **Predictions are clinically reasonable**
 - Pain & function: ~0.05
 - Rehab time: ~1 week
 - Cost: ~\$25

Clinical Heterogeneity

- The same surgery leads to very different benefit–cost profiles across patients.
- Surgery does not benefit everyone in the same way
- Different baseline clinical factors drive pain relief, functional recovery, rehabilitation time, and cost
- The need for individualized, preference-aware treatment decisions.

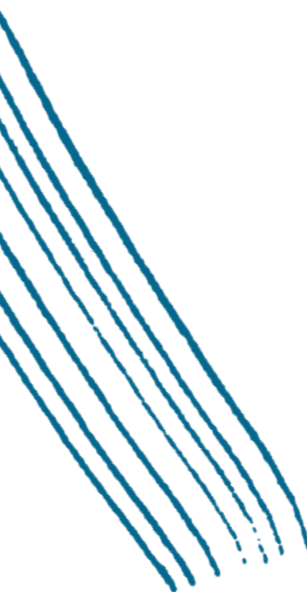
Outcome	TE Range
Pain Reduction	−0.01 to +0.53
Short-term Pain	−0.42 to −0.00
Functional Improvement	−0.01 to +0.53
Cost Impact (USD)	+\$10k to +\$30k
Rehab Time (weeks)	+3.2 to +8.7





Conclusion

*The **S-learner with XGBoost Model** can be used to compare treatment options and select the one that best fits a patient's condition and priorities.*



Thank you

